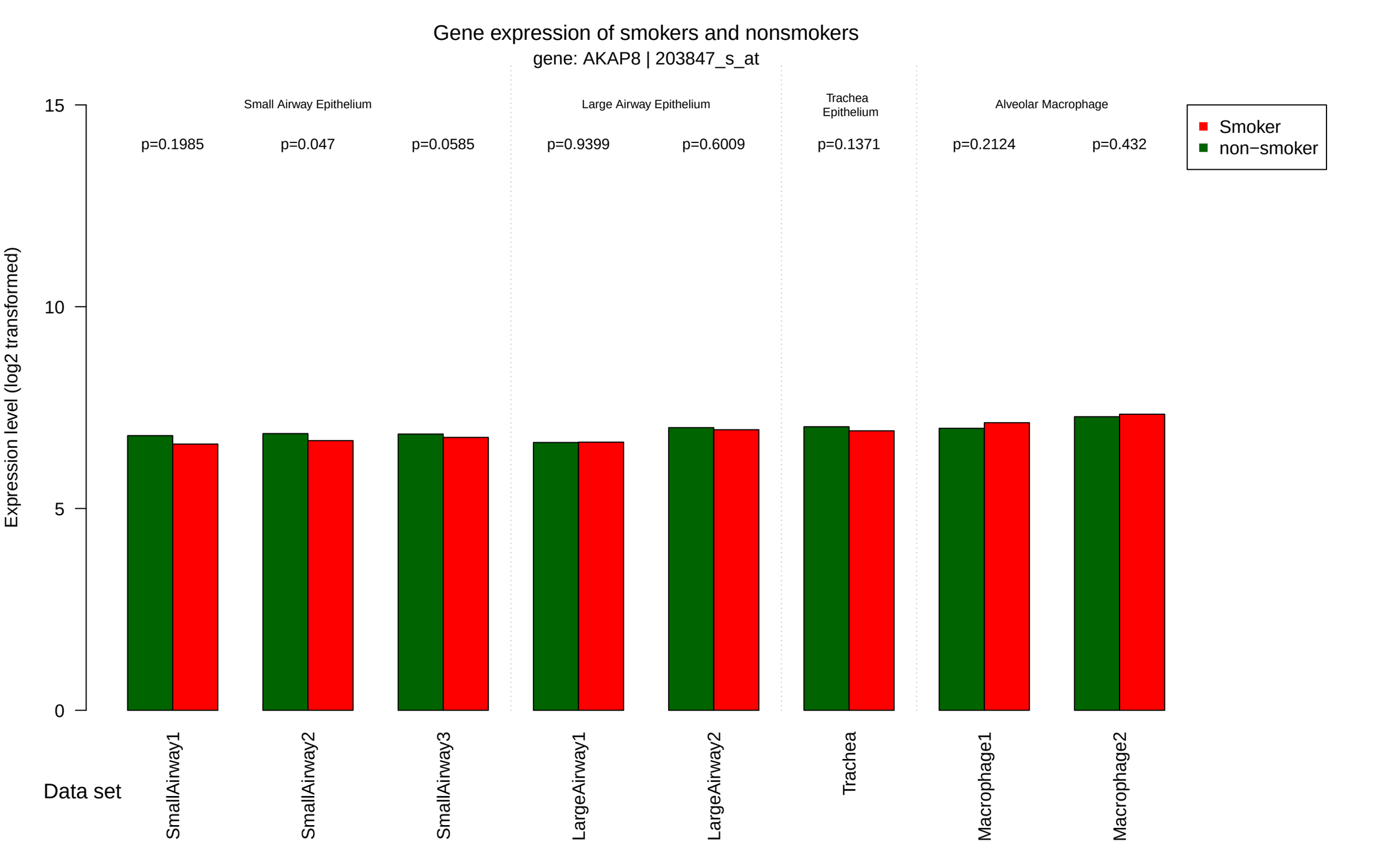
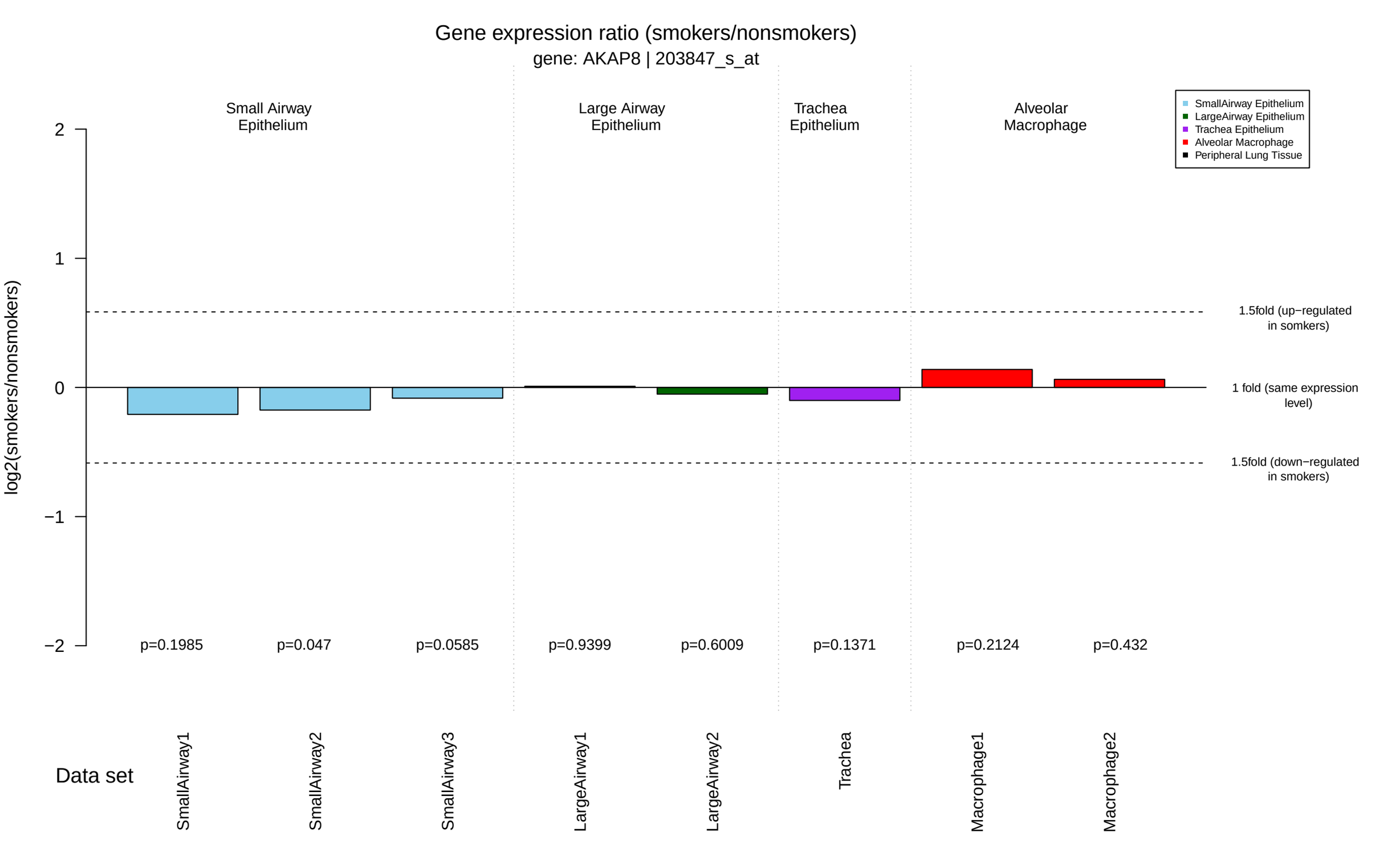
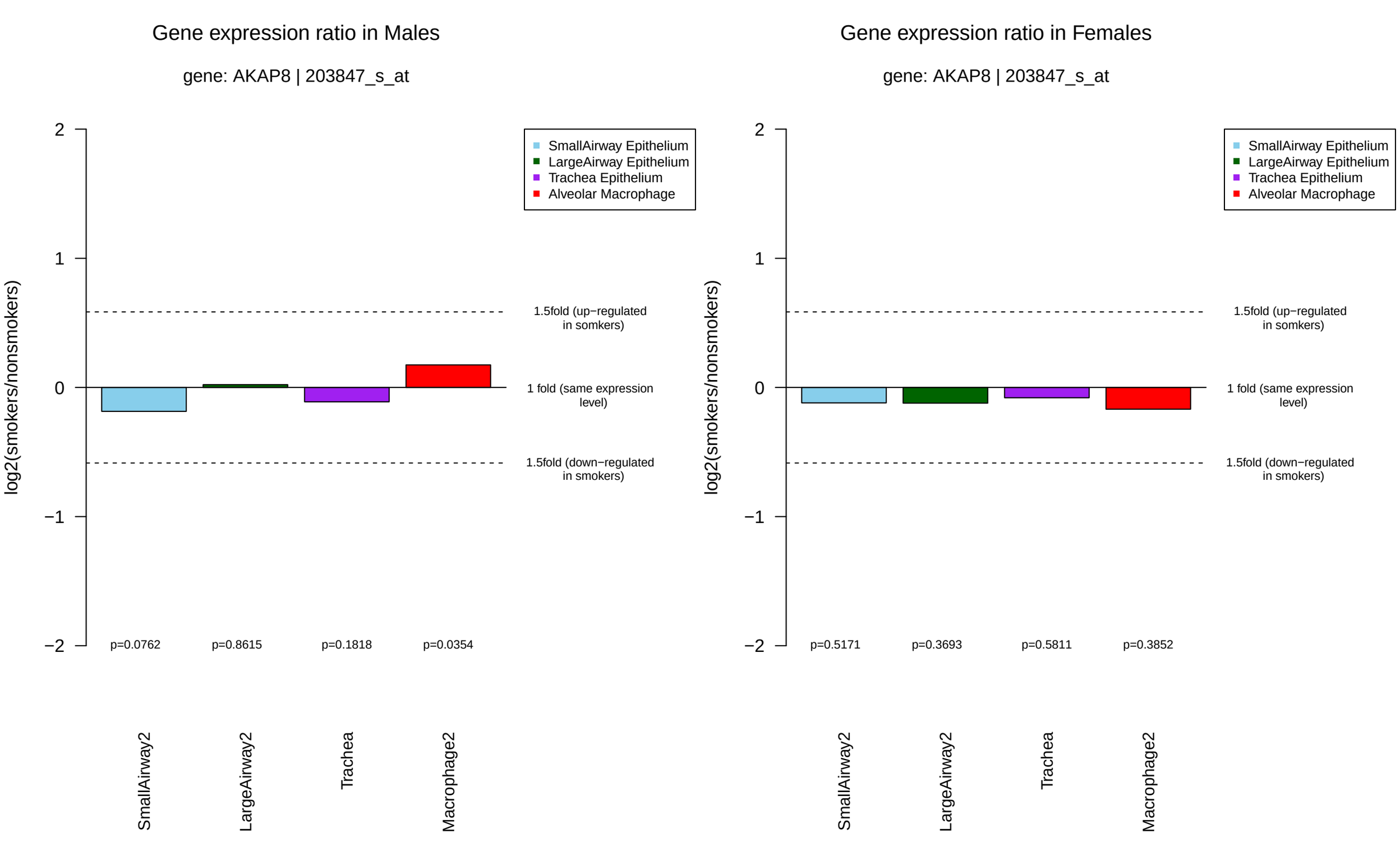
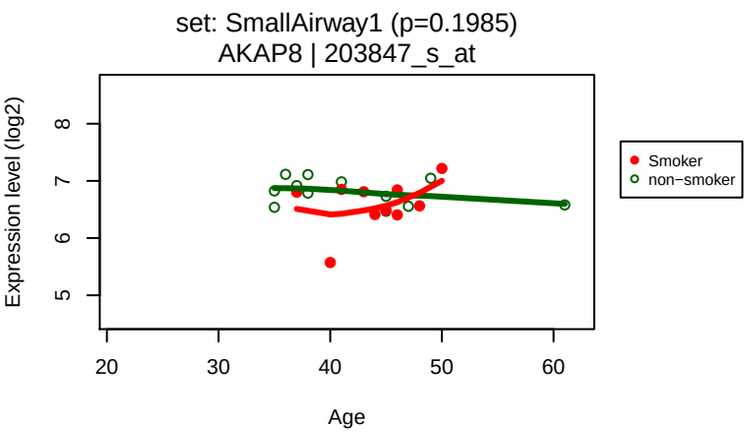
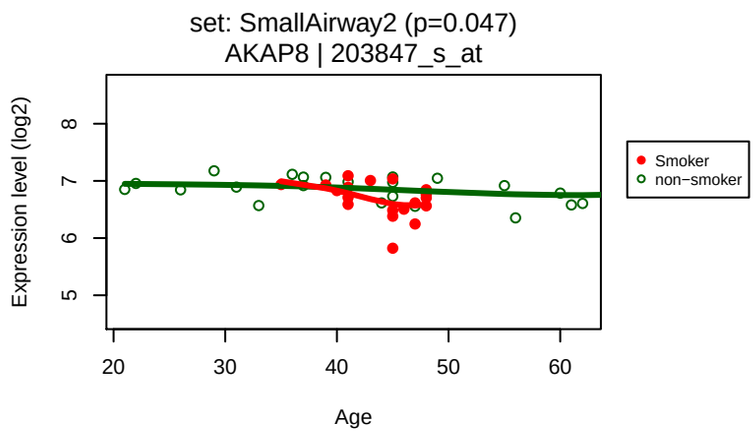
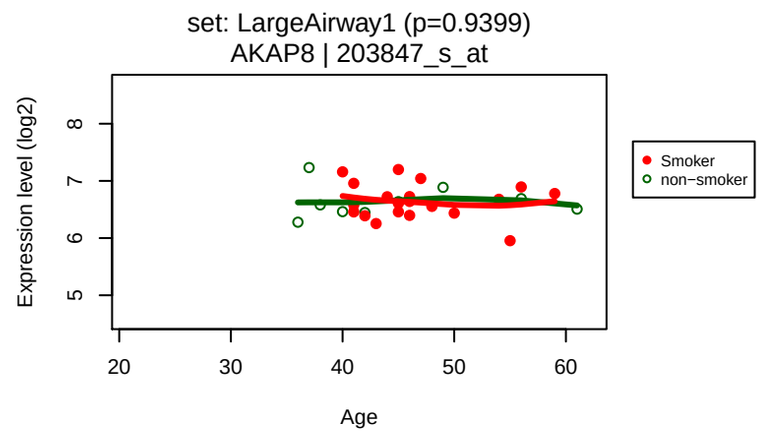

Gene expression of smokers and nonsmokers
gene: AKAP8 | 203847_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1985
p=0.047
p=0.0585
p=0.9399
p=0.6009
p=0.1371
p=0.2124
p=0.432
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

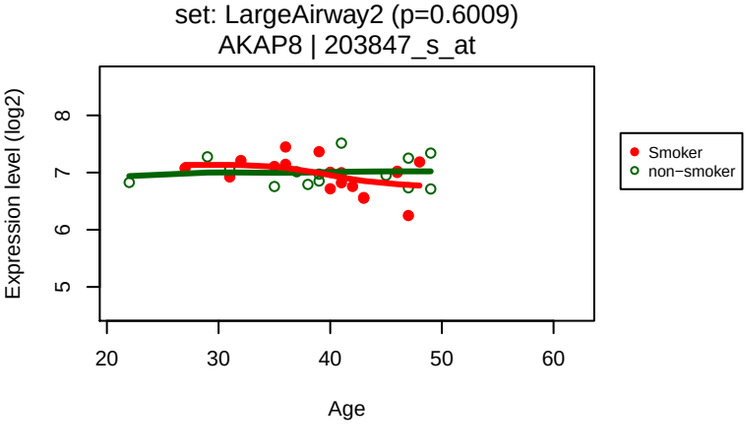

Gene expression ratio (smokers/nonsmokers)
gene: AKAP8 | 203847_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1985
p=0.047
p=0.0585
p=0.9399
p=0.6009
p=0.1371
p=0.2124
p=0.432
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

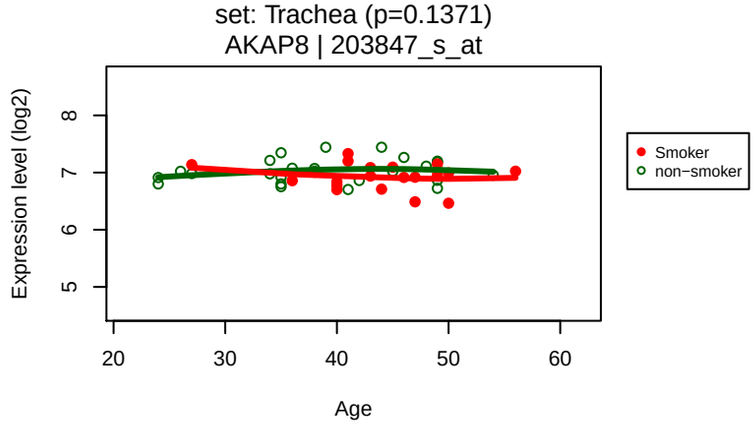

Gene expression ratio in Males
gene: AKAP8 | 203847_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0762
p=0.8615
p=0.1818
p=0.0354
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: AKAP8 | 203847_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5171
p=0.3693
p=0.5811
p=0.3852
SmallAirway2
LargeAirway2
Trachea
Macrophage2

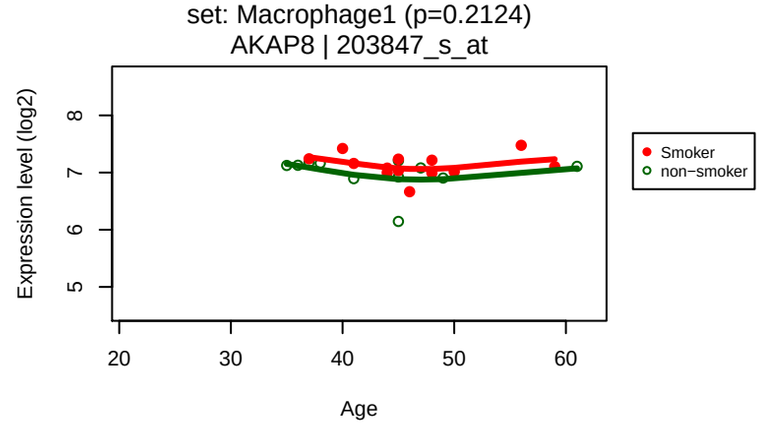

set: SmallAirway1 (p=0.1985)
AKAP8 | 203847_s_at
Expression level (log2)
Age
Smoker
non−smoker

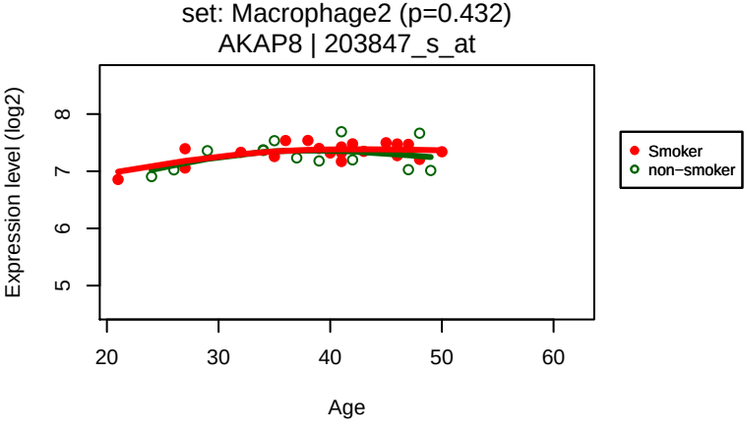

set: SmallAirway2 (p=0.047)
AKAP8 | 203847_s_at
Expression level (log2)
Age
Smoker
non−smoker

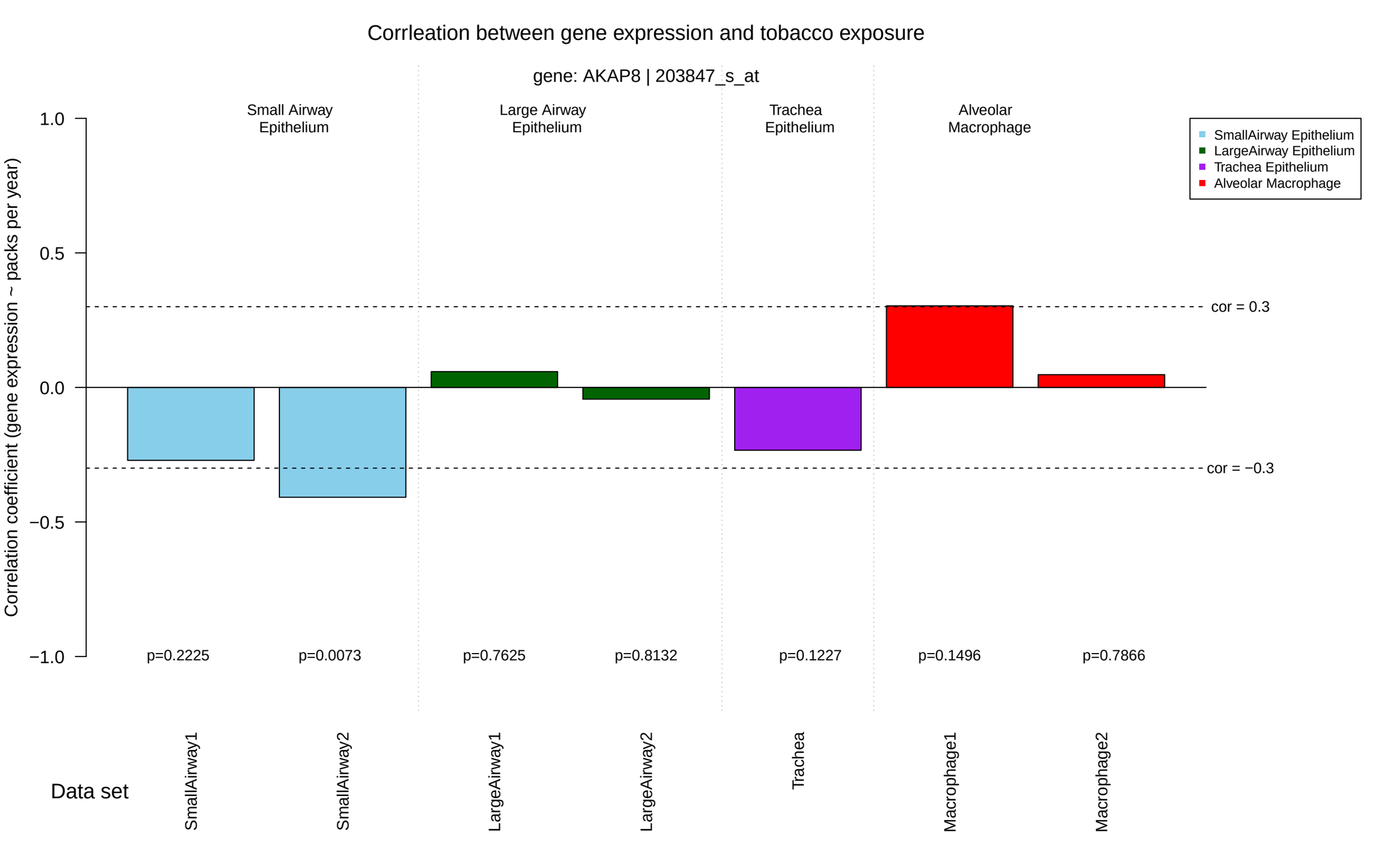

set: LargeAirway1 (p=0.9399)
AKAP8 | 203847_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6009)
AKAP8 | 203847_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1371)
AKAP8 | 203847_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2124)
AKAP8 | 203847_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.432)
AKAP8 | 203847_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: AKAP8 | 203847_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2225
p=0.0073
p=0.7625
p=0.8132
p=0.1227
p=0.1496
p=0.7866
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2